Table 1: Fitted coefficients for all three SINDy models. Each cell shows the coefficient of the row's basis function in the corresponding state equation. Dashes (—) indicate the term is absent. **Model 1** (STLSQ, λ=0.20): *best fit with hysteresis, not sparse*, R^2=0.885, 52 parameters, $\tau \in [-800\,\mathrm{kyr}, 0]$. **Model 2** (ADMM): *sparse, no hysteresis*, R^2=0.846, 25 parameters, $\tau \in [-800\,\mathrm{kyr}, 0]$. **Model 3** (STLSQ, λ=6.31): *all attractors, 2 Myr period*, R^2=0.838, 38 parameters, $\tau \in [-2\,\mathrm{Myr}, 0]$. States: $H, C, B, H_s \equiv H_{\mathrm{sed}}, T$. Input: I.

	modA**Model 1** — STLSQ (λ=0.20), R^2=0.885					modB**Model 2** — ADMM, R^2=0.846					modC**Model 3** — STLSQ (λ=6.31), R^2=0.838				
Term	$\dot{H}$	$\dot{C}$	$\dot{B}$	$\dot{H}_s$	$\dot{T}$	$\dot{H}$	$\dot{C}$	$\dot{B}$	$\dot{H}_s$	$\dot{T}$	$\dot{H}$	$\dot{C}$	$\dot{B}$	$\dot{H}_s$	$\dot{T}$
1	−809.26	349.47	−117.15	1.58	67.93	—	2.19	—	—	—	−1050.60	303.39	−161.55	15.55	35.74
H	485.74	19.17	53.36	1.86	3.33	73.40	—	−13.86	−0.15	—	201.32	—	10.06	—	—
C	−35.94	8.64	−6.52	0.51	0.38	175.51	—	2.23	0.13	—	76.43	—	23.27	—	—
B	25.40	−5.23	−17.96	—	−1.11	18.76	−7.48	−18.60	—	−1.32	105.97	—	—	—	—
H_s	31.57	15.20	6.45	−3.37	4.57	−42.55	—	—	—	—	153.34	70.47	15.11	−12.77	—
T	878.99	−459.05	151.91	—	−91.16	—	—	19.71	—	—	1093.42	−389.52	183.26	−15.46	−43.01
I	−80.61	90.25	−13.09	—	19.26	−148.12	51.45	—	—	1.05	134.41	−6.71	44.32	−15.05	−27.90
H^2	−411.16	−18.75	−66.63	−2.96	−3.75	−32.96	−20.57	—	—	−1.45	—	—	—	—	—
$T \cdot H$	−351.63	−39.79	−64.71	−1.89	−7.76	—	−3.45	−15.51	—	—	—	—	—	—	—
$C \cdot H$	—	—	—	—	—	—	—	—	—	—	−162.94	−18.41	−35.04	—	—
$B \cdot C$	—	—	—	—	—	—	—	—	—	—	−86.51	—	−19.72	—	—
$T \cdot H_s$	—	—	—	—	—	—	—	—	—	—	−162.57	−74.11	−16.02	12.98	—
$T \cdot I$	—	—	—	—	—	—	—	—	—	—	−260.51	96.20	−67.89	14.98	35.57
H^3	580.28	25.95	94.61	−1.89	5.25	—	—	—	—	—	—	—	—	—	—
H^4	−264.19	−11.96	−43.32	1.45	−2.39	16.50	6.90	0.92	—	—	—	—	—	—	—

Notes. (i) Model 1 includes polynomial nonlinearities up to H^4 plus the interaction $T \cdot H$, yielding good hysteresis reproduction at the cost of 52 parameters. (ii) Model 2 (ADMM) selects a sparser library (25 parameters) but loses the hysteresis cycle. (iii) Model 3 uses bilinear cross-terms ($C \cdot H$, $B \cdot C$, $T \cdot H_s$, $T \cdot I$) instead of high-degree polynomials, covering the full 2 Myr attractor set. $H_s \equiv H_{\mathrm{sed}}$.